Fibonacci Polynomials & *Lights Out*

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Abstract

Consider a game played on a simple graph $G = (V, E)$ where each vertex consists of clickable light. A move is clicking any $v \in V$, which toggles the on/off state of v and its neighbors. Given an initial configuration, one wins the game by finding a sequence of moves that turns off all of the lights. This game was commercially available from Tiger Electronics as *Lights Out* for G a 5×5 grid.

Let $\Phi_G : P(V) \to P(V)$ be the function which maps the vertices clicked to the vertices that change state. For G and $n \times n$ grid, let $d(n) = \dim(\ker \Phi_G)$. We show several that many values of $d(n)$ never occur. We show more general formulas for the value of $d(n)$, including a complete characterization when $n = 2^a p^b - 1$ for $a, b \geq 0$ and p a Wieferich prime.

1 Fibonacci Polynomials

Definition 1.1 (Fibonacci Polynomial)**.** *Let $F_n(x)$ be the polynomial in the ring $\mathbb{F}_2[x]$ defined recursively by*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & \textit{otherwise} \end{cases}.$$

These polynomials are called "Fibonacci" because when defined over the real numbers, $F_n(1)$ is the nth Fibonacci number.

In $\mathbb{F}_2[x]$, the coefficients of $F_n(x)$ are either 1 or 0, so we can visualize them by coloring squares black or white to represent the coefficients. For example,

$$F_6(x) = 1x^5 + 0x^4 + 0x^3 + 0x^2 + 1x + 0 = \blacksquare\square\square\square\blacksquare\square.$$

Plotting $F_1(x)$ through $F_{128}(x)$ in the same way, aligning terms of the same degree, we see a Sierpinski triangle.

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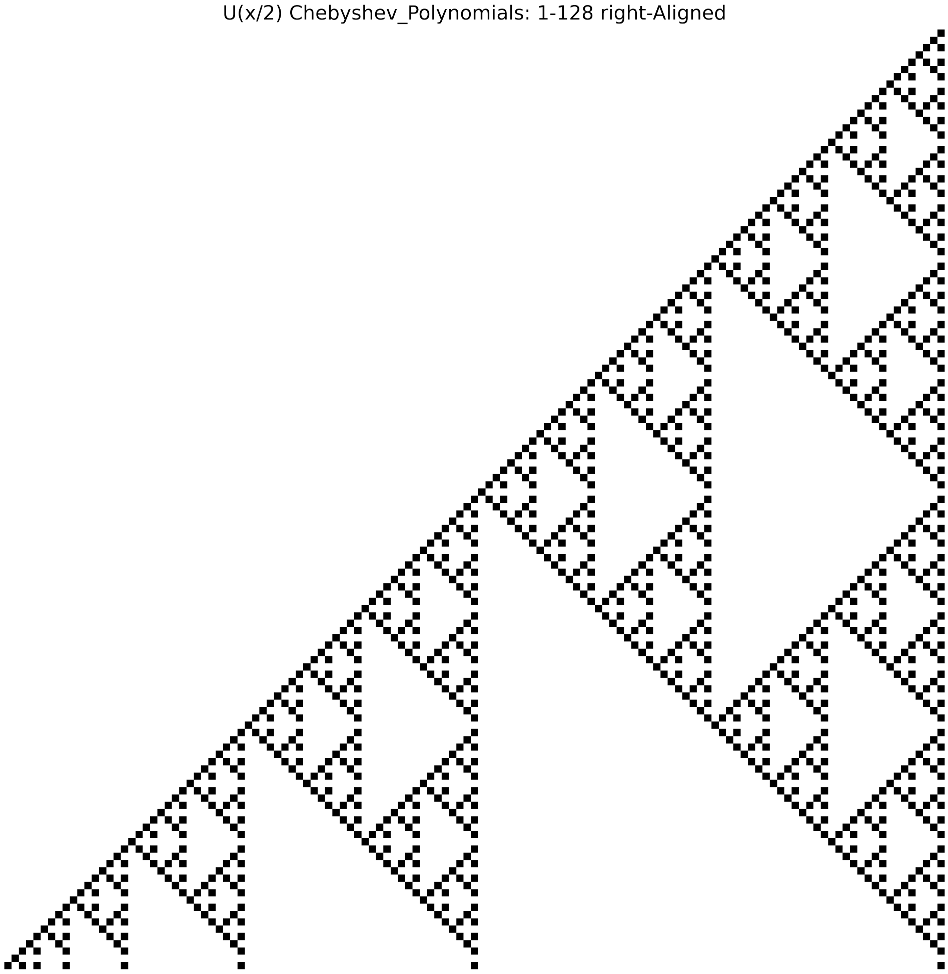

Figure 1: $F_1(x)$ through $F_{128}(x)$, right-aligned.

Using a well-known connection between the Sierpinski triangle and the parity of binomial coefficients, we may see that

$$F_n(x) = \sum_{i=0}^{n-1} \binom{n+i}{2i+1} x^i \mod 2.$$

The above construction and formula was also shown by Sutner [6]. Sutner also showed the following relationship between square *Lights Out* grids and Fibonacci Polynomials.

Lemma 1.1 (Sutner)**.** *Let $G = (V, E)$ be an $n \times m$ grid graph. Let $\Phi_G : P(V) \to P(V)$ be the function that maps which lights are clicked to which lights change state. Then*

$$\dim(\ker \Phi_G) = \deg\left(\gcd\left(F_{n+1}(x), F_{m+1}(x+1)\right)\right) = \deg\left(\gcd\left(F_{n+1}(x+1), F_{m+1}(x)\right)\right).$$

Proof. See [6] theorem 5.1. ■

Sutner and others also shows several divisibility rules for these polynomials

Lemma 1.2 (Hunziker, Machiavelo, and Park)**.** *A polynomial $\tau(x)$ in $\mathbb{F}_2[x]$ divides both $F_n(x)$ and $F_m(x)$ if and only if it divides $F_{\gcd(m,n)}(x)$. In particular,*

$$\gcd\left(F_m(x), F_n(x)\right) = F_{\gcd(m,n)}(x).$$

Proof. See [1] proposition 2.2, [6] proposition 2.1, and [2]. ■

Corollary 1.2.1. *The following are true:*

(i) x *divides* $F_n(x)$ *if and only if* $n \equiv 0 \mod 2$.

(ii) $x+1$ *divides* $F_n(x+1)$ *if and only if* $n \equiv 0 \mod 2$.

(iii) $x+1$ *divides* $F_n(x)$ *if and only if* $n \equiv 0 \mod 3$.

(iv) x *divides* $F_n(x+1)$ *if and only if* $n \equiv 0 \mod 3$.

Proof. Notice,

(i) We find that $f_2(x) = x$ is the smallest Fibonacci polynomial divisible by x, so we apply lemma 1.2 to get the desired result.

(ii) Follows from (i) by substituting $x+1$ for x.

(iii) We find that $f_3(x) = (x+1)^2$ is the smallest Fibonacci polynomial divisible by $x+1$, so we apply lemma 1.2 to get the desired result.

(iv) Follows from (iii) by substituting $x+1$ for x.

■

Lemma 1.3 (Hunziker, Machiavelo, and Park)**.** *If n is odd, then $F_n(x)$ is the square of a square-free polynomial.*

Proof. See [1] proposition 2.11 and [6] Theorem 2.1 claim 2. ■

Corollary 1.3.1. *If n is odd, then every root of $F_n(x)$ has multiplicity 2.*

Proof. Let n be odd and α a root of $F_n(x)$. By 1.3, there exists $\tau_n \in \mathbb{F}_2[x]$ such that

$$F_n(x) = \tau_n(x)^2,$$

and $\tau(x)$ is square-free. Since α is a root of $F_n(x)$, it must also be a root of $\tau_n(x)$ and thus has multiplicity 2. ■

We can now prove some of our own facts about these polynomials and their GCDs, particularly on square grids.

Lemma 1.4. *Let $d(n) = \dim(\ker \Phi_G)$ for G an $n \times n$ grid graph. Then $d(n)$ is always even.*

Proof. Let

$$G_n(x) = \gcd\left(F_n(x), F_n(x+1)\right).$$

By lemma 1.1, we know that $d(n) = \deg G_{n+1}(x)$ and $G_n(x) = G_n(x+1)$.

$y = x(x+1)$ is also unchanged by this $x \mapsto x+1$ translation. It is well-known that every polynomial $P(x)$ in $\mathbb{F}_2[x]$ can be uniquely written as

$$P(x) = A(y) + xB(y) \qquad A, B \in \mathbb{F}_2[y].$$

We can then determine exactly when P is also unchanged by translation.

$$\begin{aligned} P(x+1) &= P(x) \\ A(y) + (x+1)B(y) &= A(y) + xB(y) \\ B(y) &= 0. \end{aligned}$$

In other words, $P(x)$ in unchanged by translation when P is a polynomial in $x^2 + x$.

Since G_n is also unchanged by translation, there exists $H_n \in \mathbb{F}_2[x]$ such that

$$G_n(x) = H_n(x^2 + x).$$

So,

$$d(n) = \deg G_{n+1}(x) = 2 \deg H_{n+1}(x)$$

and thus $d(n)$ is even. ∎

Corollary 1.4.1. *If n is even, then $d(n)$ is divisible by 4.*

Proof. If n is even, then $n+1$ is odd. Then by 1.3, $F_{n+1}(x)$ and $F_{n+1}(x+1)$ are squares of square-free polynomials $\tau(x)$ and $\tau(x+1)$.

$$\begin{aligned} G_{n+1}(x) &= \gcd\left(F_{n+1}(x), F_{n+1}(x+1)\right) \\ &= \gcd\left(\tau_{n+1}(x)^2, \tau_{n+1}(x+1)^2\right) \\ &= \gcd\left(\tau_{n+1}(x), \tau_{n+1}(x+1)\right)^2 \\ \sqrt{G_{n+1}(x)} &= \gcd\left(\tau_{n+1}(x), \tau_{n+1}(x+1)\right) \end{aligned}$$

We can see from the right-hand side of the equation that this function is also unaffected by translation. So, by lemma 1.4, there exists $H_{n+1} \in \mathbb{F}_2[x]$ such that

$$\begin{aligned} \sqrt{G_{n+1}(x)} &= H_{n+1}(x^2 + x) \\ G_{n+1}(x) &= H_{n+1}(x^2 + x)^2. \end{aligned}$$

So,

$$d(n) = \deg G_{n+1}(x) = 4 \deg H_{n+1}(x),$$

as desired. ∎

Note that the implication does not hold in the other direction: $d(n)$ also be divisible by 4 when n is odd. For example, $d(9) = 8$.

Lemma 1.5. *Let $x = t + t^{-1}$. Then for $t \neq 0$,*

$$xF_n(x) = t^n + t^{-n}.$$

Proof. Define $G_n(t) = t^{n+1} + t^{-(n+1)}$. Then

$$\begin{aligned} xG_{n-1}(t) + G_{n-2}(t) &= (t + t^{-1})(t^n + t^{-n}) + (t^{n-1} + t^{-(n-1)}) \\ &= t^{n+1} + t^{-(n-1)} + t^{n-1} + t^{-(n+1)} + t^{n-1} + t^{-(n-1)} \\ &= t^{n+1} + t^{-(n+1)} \\ &= G_n. \end{aligned}$$

Thus we see that $G_n(t)$ satisfies the same recurrence relation as $F_n(x)$. $xF_n(x)$ also satisfies the same recurrence relation.

$$\begin{aligned} xF_n(x) &= x(xF_{n-1}(x) + F_{n-2}(x)) \\ &= x(xF_{n-1}(x)) + xF_{n-2}(x). \end{aligned}$$

Now we compute $xF_1(x)$ and $xF_2(x)$ in terms of t and G:

$$\begin{aligned} xF_1(x) &= x \\ &= t + t^{-1} \\ &= G_0(t), \end{aligned}$$

and

$$\begin{aligned} xF_2(x) &= x^2 \\ &= (t + t^{-1})^2 \\ &= t^2 + t^{-2} \\ &= G_1(t). \end{aligned}$$

Since the first two terms match and they satisfy the same recurrence relation then we've shown by induction that for $t \neq 0$,

$$xF_n(x) = G_{n-1}(t) = t^n + t^{-n}.$$

$\blacksquare$

This was also proven in [1], who also prove the following identities.

Lemma 1.6 (Hunziker, Machiavelo, and Park)**.** *Let $n = 2^k b$ where $b, k \in \mathbb{N}$. Then*

$$F_n(x) = x^{2^k - 1} F_b(x)^{2^k}.$$

Proof. See [1] lemma 2.6. $\blacksquare$

Lemma 1.7. *The distinct roots of $F_n(x)$ are $\{\zeta + \zeta^{-1} : \zeta^n = 1, \zeta \neq 1\}$, plus 0 when n is even.*

Proof. Notice that $F_n(0) = F_{n-2}(0)$, so

$$F_n(0) = \begin{cases} 1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}.$$

Thus 0 is a root of $F_n(x)$ whenever n is even.

Let α be a non-zero root of F_n. Choose ζ satisfying

$$\zeta^2 + \alpha\zeta + 1 = 0.$$

Notice that ζ cannot be 0. Dividing by ζ,

$$\begin{aligned} \zeta + \alpha + \zeta^{-1} &= 0 \\ \alpha &= \zeta + \zeta^{-1}. \end{aligned}$$

By, Lemma 1.5,

$$\begin{aligned} 0 &= \alpha F_n(\alpha) \\ &= \zeta^n + \zeta^{-n} \end{aligned}$$

Since a field has no nonzero nilpotents, we may multiply by ζ^n.

$$\begin{aligned} 0 &= \zeta^{2n} + 1 \\ &= (\zeta^n + 1)^2 \\ &= \zeta^n + 1 \\ \zeta^n &= 1. \end{aligned}$$

Thus the order of ζ divides n. Notice too that $\zeta \neq 1$; otherwise, $\alpha = 1 + 1^{-1} = 0$, which we already showed as not possible.

Conversely, if $\zeta^n = 1$ and $\zeta \neq 1$, then let $\alpha = \zeta + \zeta^{-1}$. Then we see $\alpha \neq 0$ and

$$\alpha F_n(\alpha) = \zeta^n + \zeta^{-n} = 1 + 1^{-1} = 0.$$

Thus, α is a root of F_n. ■

2 Even Parity Covers

$\ker \Phi_G$ may also be understood via parity covers and dominating sets.

Definition 2.1. *An even dominating set of a graph $G = (V, E)$ is a subset $V' \subseteq V$ such that every vertex contains an even number of vertices from V' in its neighborhood.*

Odd dominating sets can be analogously defined. Sutner specifically studied odd dominating sets as the solution to *Lights Out* when all lights are initially on. He showed that such an odd dominating set always exists for any finite graph [5].

Lemma 2.1. *$V' \in \ker \Phi_G$ if and only if V' is an even dominating set of G.*

Proof. Let $V' \in \ker \Phi_G$. Let $N(v)$ be the neighborhood of vertex v: v and its immediate neighbors. Then for every $v \in V$,

$$|N(v) \cap V'| \text{ is even.}$$

Thus, V' is an even dominating set of G. We can apply the same steps in reverse to show the converse. ■

Lemma 2.2. *Φ_G is linear with respect to symmetric difference. That is, for all $V_1, V_2 \subseteq V$,*

$$\Phi_G(V_1 \triangle V_2) = \Phi_G(V_1) \triangle \Phi_G(V_2).$$

Proof.

$$\begin{aligned}
\Phi_G(V_1 \triangle V_2) &= \{v \in V : |N(v) \cap (V_1 \triangle V_2)| \text{ is odd}\} \\
&= (\{v \in V : |N(v) \cap V_1| \text{ is odd}\} \cup \{v \in V : |N(v) \cap V_2| \text{ is odd}\}) \setminus \{v \in V : |N(v) \cap (V_1 \cap V_2)| \text{ is odd}\} \\
&= \{v \in V : |N(v) \cap V_1| \text{ is odd}\} \triangle \{v \in V : |N(v) \cap V_2| \text{ is odd}\} \\
&= \Phi_G(V_1) \triangle \Phi_G(V_2)
\end{aligned}$$

■

Corollary 2.2.1. *If V_1 and V_2 have the same image under Φ_G if and only if $V_1 \triangle V_2 \in \ker \Phi_G$.*

Proof. Assume that V_1 and V_2 have the same image under Φ_G. Then

$$\begin{aligned}
\Phi_G(V_1 \triangle V_2) &= \Phi_G(V_1) \triangle \Phi(V_2) \\
&= \Phi_G(V_1) \triangle \Phi_G(V_2) \\
&= \emptyset.
\end{aligned}$$

Thus, $V_1 \triangle V_2 \in \ker \Phi_G$. Now assume $V_1 \triangle V_2 \in \ker \Phi_G$. Then

$$\begin{aligned}
\Phi_G(V_1) &= \Phi_G(V_1) \triangle \emptyset \\
&= \Phi_G(V_1) \triangle \Phi_G(V_1 \triangle V_2) \\
&= \Phi_G(V_1) \triangle \Phi_G(V_1) \triangle \Phi_G(V_2) \\
&= \Phi_G(V_2).
\end{aligned}$$

Thus V_1 and V_2 have the same image under Φ_G. ■

We may now use our geometric understanding of $\ker \Phi_G$ as even parity covers to construct a lower bound on $d(n)$.

Theorem 2.1. *For all $n, k \in \mathbb{N}$,*

$$d(nk-1) \geq d(n-1).$$

Proof. Let G_L be an $nk-1 \times nk-1$ grid and G_ℓ be an $n-1 \times n-1$ grid. Since $nk-1 = k(n-1)+(k-1)$, an G may be broken into a $k \times k$ grid of $n-1 \times n-1$ grids, each separated by a horizontal or vertical strip of size 1.

Let Q be an even dominating set of G_ℓ. Let R be Q tiles k times horizontally and vertically, with size 1 separators between each tile, onto G_L where each tile is the reflection across its neighboring separators, as shown below.

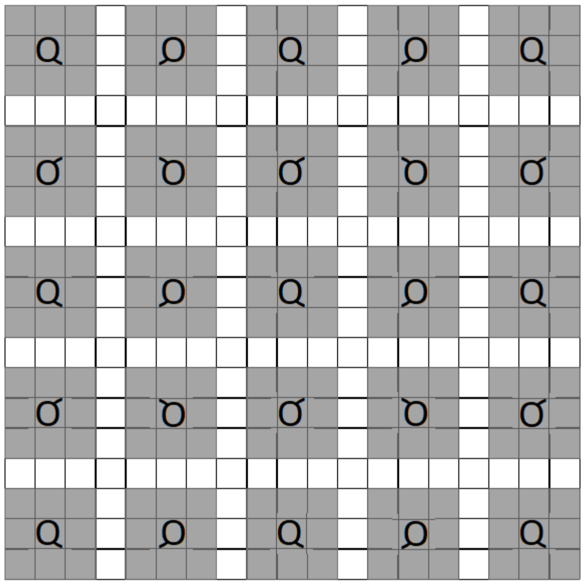

Figure 2: Pattern Q tiled 5 times.

Consider a vertex v in G_L. There are 3 cases.

1. v is within a tile. Then $|N(v) \cap R|$ is even because Q being is an even parity cover.
2. v is on a separator between two tiles. Then only two elements in $N(v)$, the vertices within a tile, are also in R. Because neighboring tiles are reflections of each other across separators, these two vertices are either both in R or both not in R. In either case, $|N(v) \cap R|$ is even.
3. v is in the intersection of two separators. Then all of $N(v)$ is in a separator and thus not in R, so $|N(v) \cap R| = 0$, which is even.

Since for every vertex v of G_L, $|N(v) \cap R|$ is even, so R is an even dominating set of G_L. Thus, for every even dominating set of G_ℓ, there exists an even dominating set of G_L obtained through tiling. This tiling function is also injective because we can recover Q by restricting R to just the top left tile. Thus,

$$d(nk-1) \geq d(n-1),$$

as desired. ∎

Sutner shows a similar result [4]. There is also a simple proof in terms of Fibonacci polynomials that also provides a divisibility relationship.

Proof. Since n divides nk, lemma 1.2 tells us $F_n(x)$ divides $F_{nk}(x)$. So,

$$\gcd(F_n(x), F_n(x+1)) \mid \gcd(F_{nk}(x), F_{nk}(x+1)).$$

So we may write

$$\gcd(F_{nk}(x), F_{nk}(x+1)) = \gcd(F_n(x), F_n(x+1))Q(x),$$

for some $Q(x) \in \mathbb{F}_2[x]$. Taking the degrees, we obtain $d(nk-1) = d(n-1) + \deg Q(x)$, and so the desired inequality follows. ■

Note that although the smaller GCD polynomial divides the larger, it's not the case that $d(n-1)$ divides $d(nk-1)$. For example, if $n = 5$ and $k = 6$, then $d(n-1) = d(4) = 4$, which does not divide $d(nk-1) = d(29) = 10$.

3 Number Theory

We may also understand $d(n)$ and the Fibonacci polynomials with some results from number theory.

Lemma 3.1 (Ore)**.** *Let a, b, c, and d be integers. Let (a, b) denote $\gcd(a, b)$. Then*

$$(ab, cd) = (a, c)(b, d)\left(\frac{a}{(a, c)}, \frac{d}{(b, d)}\right)\left(\frac{c}{(a, c)}, \frac{b}{(b, d)}\right).$$

Proof. See [7]. ■

Although Ore's result deals specifically with integers, both the integers and $\mathbb{F}_2[x]$ are Euclidean domains, so the result will still hold for our purposes.

Theorem 3.1. *For all $n \in \mathbb{N}$,*

$$d(2n+1) = 2d(n) + \delta_n, \qquad \delta_n = \begin{cases} 2 & 3 \mid n+1 \\ 0 & \textit{otherwise} \end{cases}.$$

Proof. Let (a, b) denote $\gcd(a, b)$. Applying lemmas 1.6 and 3.1,

$$\begin{aligned}
d(2n+1) &= \deg(F_{2n+2}(x), F_{2n+2}(x+1)) \\
&= \deg\left(xF_{n+1}(x)^2, (x+1)F_{n+1}(x+1)^2\right) \\
&= \deg(x, x+1)\left(F_{n+1}(x)^2, F_{n+1}(x+1)^2\right)\left(\frac{x+1}{(x, x+1)}, \frac{F_{n+1}(x)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)}\right)\left(\frac{x}{(x, x+1)}, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)}\right) \\
&= \deg(F_{n+1}(x), F_{n+1}(x+1))^2\left(x+1, \frac{F_{n+1}(x)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2}\right)\left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2}\right) \\
&= 2d(n) + 2\deg\left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2}\right)
\end{aligned}$$

Thus, we have that

$$\delta_n = 2\deg\left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2}\right).$$

So, δ_n is 2 exactly when $F_{n+1}(x)$ can be divided without remainder by $x+1$ more times than $F_{n+1}(x+1)$. When $n+1$ is not divisible by 3, corollary 1.2.1 tells us that $F_{n+1}(x)$ is not divisible by $x+1$, so $\delta_n = 0$. When $n+1$ is divisible by 3, let $n+1 = b2^k$ for $b, k \geq 0$ and b odd. Then b must be divisible by 3. Applying Lemma 1.6,

$$F_{n+1}(x) = x^{2^k-1} F_b(x)^{2^k} \qquad F_{n+1}(x+1) = (x+1)^{2^k-1} F_b(x+1)^{2^k}.$$

Since b is an odd multiple of 3, corollary 1.2.1 tells us that $x+1$ divides $F_b(x)$, but $x+1$ does not divide $F_b(x+1)$. So,

$$F_{n+1}(x) = x^{2^k-1}(x+1)^{2^k} g(x)^{2^k} \qquad F_{n+1}(x+1) = (x+1)^{2^k-1} x^{2^k} g(x+1)^{2^k},$$

for some $g(x) \in \mathbb{F}_2[x]$, where $g(x)$ and $g(x+1)$ are both divisible by neither x nor $x+1$. So, we see that $F_{n+1}(x)$ can be divided without remainder by $x+1$ more times than $f_{n+1}(x+1)$, so $\delta_n = 2$. ■

This result resolves a conjecture of Sutner's [5].

We can repeatedly apply theorem 3.1 to obtain explicit formulas for $d(n)$.

Corollary 3.1.1. *Write odd n as $n+1 = 2^s b$ for $s \geq 0$ and b odd. Then*

$$d(n) = 2^s d(b-1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \textit{otherwise} \end{cases}.$$

Proof. By induction on s. For $s = 0$,

$$\begin{aligned} n+1 &= 2^0 b \\ n &= b-1 \\ d(n) &= d(b-1) \\ &= 2^0 d(b-1) + \begin{cases} 2(2^0 - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}, \end{aligned}$$

as desired. Assume that our claim holds for $s \geq 0$. Then for $n+1 = 2^{s+1} b$,

$$\begin{aligned} d(n) &= d\left(2^{s+1} b - 1\right) \\ &= d\left(2\left(2^s b - 1\right) + 1\right) \\ &= 2d\left(2^s b - 1\right) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\ &= 2\left(2^s d(b-1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}\right) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\ &= 2^{s+1} d(b-1) + \begin{cases} 2(2^{s+1} - 1) - 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} + \begin{cases} 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \\ &= 2^{s+1} d(b-1) + \begin{cases} 2(2^{s+1} - 1) & 3 \mid b \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

as desired. ■

We can also use theorem 3.1 to narrow down more when $d(n) = 2$.

Corollary 3.1.2. *If $d(n) = 2$, then $n \equiv -1 \pmod 6$.*

Proof. Since 2 is not divisible by 4, corollary 1.4.1 tells us that n must be odd. So, we can write $n = 2^k b - 1$ for b odd and $k \geq 1$. Using corollary 3.1.1, we can define

$$D_i = d(2^i b - 1) = \begin{cases} 2^i d(b-1) + 2(2^i - 1) & 3 \mid b \\ 2^i d(b-1) & \text{otherwise} \end{cases}.$$

If $3 \nmid b$, then we must have $D_k = 2 = 2^k d(b-1)$. However, corollary 1.4.1 tells us that $d(b-1)$ is divisible by 4, which cannot be the case.

If $3 \mid b$, then we must have $D_k = 2 = 2^k d(b-1) + 2(2^k - 1)$. If $k \geq 2$, then this expression is greater than 2, which cannot be the case. Thus, we must have $k = 1$ and $D_1 = 2 = 2d(b-1) + 2$, which simplifies to $d(b-1) = 0$. Thus we may write $b = 3j$.

Solving for n, we find

$$\begin{aligned} n &= 2^k b - 1 \\ &= 2b - 1 \\ &= 6j - 1, \end{aligned}$$

so $n \equiv -1 \pmod 6$, as desired. ∎

We can now use this recurrence result to prove the impossibility of certain values of $d(n)$.

Theorem 3.2. *The smallest even k such that for all $n \in \mathbb{N}$, $d(n) \neq k$ is $k = 26$.*

Proof. Recall from lemma 1.4 that $d(n)$ is always even. Through calculation, we can find that

$$\begin{array}{cccccc} d(5) = 2 & d(4) = 4 & d(11) = 6 & d(16) = 8 & d(29) = 10 & d(84) = 12 \\ d(23) = 14 & d(19) = 16 & d(101) = 18 & d(30) = 20 & d(59) = 22 & d(62) = 24. \end{array}$$

We will show than $d(n) = 26$ is impossible.

Assume for the sake of contradiction that there exists $n \in \mathbb{N}$ such that $d(n) = 26$. Since 26 is not divisible by 4, corollary 1.4.1 that n must be odd. So, we may write $n = 2m + 1$. By theorem 3.1,

$$26 = d(n) = d(2m+1) = 2d(m) + \begin{cases} 2 & 3 \mid m+1 \\ 0 & \text{otherwise} \end{cases}.$$

Since lemma 1.4 tells us that $d(m) = 13$ is not possible, we must have $d(m) = 12$ and $m + 1$ is divisible by 3. Let $m + 1 = 2^s b$ for odd b and $s \geq 0$. By corollary 3.1.1,

$$d(m) = 2^s d(b-1) + 2(2^s - 1) = 12.$$

The only possible integer solutions here are $d(b-1) = 12$ and $s = 0$ or $d(b-1) = 5$ and $s = 1$. Since lemma 1.4 tells us that $d(b-1) = 5$ is impossible, the only remaining possibility is that $d(m) = d(b-1) = 12$ and $s = 0$. Thus, m must be even.

We know from lemma 1.1 that

$$d(m) = \deg\left(\gcd\left(F_{m+1}(x), F_{m+1}(x+1)\right)\right).$$

Since $m + 1$ is odd lemma 1.3 tells us that there exists some square-free $R_{m+1}(x)$ such that

$$F_{m+1}(x) = R_{m+1}(x)^2.$$

Define $H_{m+1}(x)$

$$H_{m+1}(x) = \gcd\left(R_{m+1}(x), R_{m+1}(x+1)\right) \qquad \gcd\left(F_{m+1}(x), F_{m+1}(x+1)\right) = H(x)^2.$$

Since $R_{m+1}(x)$ is square-free, then $H_{m+1}(x)$ is too. Since $d(m) = 12$, $\deg H_{m+1}(x) = 6$. Notice too that $H_{m+1}(x) = H_{m+1}(x+1)$, so our work in lemma 1.5 tells us that for $y = x^2 + x$, there exists square-free a cubic polynomial $h_{m+1}(y) \in \mathbb{F}_2[y]$ such that

$$H_{m+1}(x) = h_{m+1}(y).$$

There are only three possible square-free cubic polynomials in $\mathbb{F}_2[y]$ that are also 1 at $y = 0$:

1. $C_1(y) = y^3 + 1$
2. $C_2(y) = y^3 + y + 1$
3. $C_3(y) = y^3 + y^2 + 1$.

We will handle these case-by-case.

Define the "rank" of a polynomial P to be the smallest integer r such that P divides F_r. Calculating the rank of our three candidates for $h_{m+1}(x)$,

$$\begin{aligned} \operatorname{rank}\left((x^2+x)^3+1\right) &= 85 \\ \operatorname{rank}\left((x^2+x)^3+(x^2+x)+1\right) &= 63 \\ \operatorname{rank}\left((x^2+x)^3+(x^2+x)^2+1\right) &= 63 \end{aligned}$$

If either rank-63 polynomials C_2 or C_3 divides $R_{m+1}(x)$, then 63 divides $m+1$. However, by definition of rank, both polynomials would divide $R_{m+1}(x)$. C_2 and C_3 are coprime and each of degree 6 in x, so $\deg H_{m+1}(x) = \deg C_2(x) + \deg C_3(x) = 12$, contradicting our assumption that the degree was 6. If C_1 R_{m+1}, then 85 divides $m+1$. Combining this with the fact that 3 divides $m+1$ we showed earlier, we see that 255 must divide $m+1$.

Define

$$P(x) = x^4 + x^3 + 1 \qquad Q(x) = x^4 + x^3 + x^2 + x + 1.$$

P and Q are coprime to each other and C_1. The ranks of $P(x)$ and $Q(x)$ are 15 and 17 respectively. Since both 15 and 17 divide 255, and 255 divides $m+1$, then both $P(x)$ and $Q(x)$ must divide $H_{m+1}(x)$. Therefore,

$$\deg H_{m+1}(x) \geq \deg C_1(x) + \deg P(x) + \deg Q(x) = 14,$$

which contradicts our assumption that the degree was 6.

Thus, there is no $3 \mid m+1$ such that $d(m) = 12$ and therefore no n such that $d(n) = 26$. ■

Now that we know that one impossible even value of $d(n)$ exists, we can show an entire family of impossible even values also exists.

Corollary 3.1.3. *Suppose $a \in \mathbb{N}$ is even and an impossible value of $d(n)$. Then $2a+2$ is also an impossible value of $d(n)$.*

Proof. Suppose for the sake of contradiction that there exists n such that $d(n) = 2a+2$. Since $2a+2$ is not divisible by 4, then corollary 1.4.1 tells us that n must be odd, so we may write $n = 2m+1$. So, by theorem 3.1,

$$2a+2 = d(n) = d(2m+1) = 2d(m) + \delta_m \qquad \delta_m = \begin{cases} 2 & 3 \mid m+1 \\ 0 & \text{otherwise} \end{cases}.$$

If $\delta_m = 0$, then $2d(m)$ is divisible by 4, which cannot be the case since we know the value is $2a+2$. If $\delta_m = 2$, then $d(m) = a$, contradicting the impossibility of a as an output of d. ■

Expanding the recurrence, if a is an impossible value, then $2^t(a+2) - 2$ is also impossible for all $t \geq 0$. In other words, 26, 54, 110, 222, 446, ... are also all impossible values of $d(n)$.

4 Finite Fields

Although we have already been working in $\mathbb{F}_2$, we will now more directly use results from the study of finite fields.

Lemma 4.1. *Let p be an odd prime. Define $h = \operatorname{ord}_p(2)$ and $c = v_p(2^h - 1)$, where $v_p(a)$ is the p-adic valuation of a. Then for every $j \geq 1$,*

$$\operatorname{ord}_{p^j}(2) = hp^{\max(0, j-c)}.$$

Proof. Suppose $2^r \equiv 1 \pmod{p^j}$. Then h divides r, so we may write $r = ht$. By the Lifting the Exponent Lemma,

$$\begin{aligned} v_p(2^r - 1) &= v_p(2^{ht} - 1) \\ &= v_p(2^h - 1) + v_p(t) \\ &= c + v_p(t). \end{aligned}$$

So, $p^j \mid 2^{ht} - 1$ exactly when $c + v_p(t) \geq j$. The smallest possible t satisfying this inequality is $t = \max(0, j - c)$, so

$$\operatorname{ord}_{p^j}(2) = hp^{\max(0, j-c)},$$

as desired. ∎

Corollary 4.1.1. *Let p be an odd prime. If p isn't a Wieferich prime, then*

$$\operatorname{ord}_{p^j}(2) = \operatorname{ord}_p(2)\, p^{j-1}.$$

Proof. Lemma 4.1 tells us that

$$\operatorname{ord}_{p^j}(2) = \operatorname{ord}_p(2)\, p^{\max(0, j-c)}, \qquad c = v_p(2^{\operatorname{ord}_p(2)} - 1).$$

Thus it is sufficient to show that $c = 1$. By definition of $\operatorname{ord}_p(2)$, $c \geq 1$. Since p is not a Wieferich prime, for all $t \geq 2$,

$$p^t \nmid 2^{\operatorname{ord}_2(p)} - 1,$$

so $c \not\geq 2$ and therefore must be exactly 1, as desired. ∎

Definition 4.1. *Let b be coprime to n. Then "signed order" of a modulo n is*

$$\rho_b(n) = \min\{r > 0 : b^r \equiv \pm 1 \pmod{n}\}.$$

This is also sometimes called the "index" of " ± 1 relative to b, written $\operatorname{ind}_b(\pm 1)$.

Lemma 4.2. *Let $H_j = \operatorname{ord}_{p^j}(2)$. Then*

$$\rho_2(p^j) = \begin{cases} H_j/2 & H_j \text{ is even} \\ H_j & H_j \text{ is odd} \end{cases}.$$

Moreover, when H_j is even,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Proof. Assume H_j is even. Then by definition of H_j,

$$\left(2^{H_j/2}\right)^2 \equiv 1 \pmod{p^j}.$$

Thus the only solutions are $2^{H_j/2} \equiv \pm 1 \pmod{p^j}$. However, 1 cannot be a solution; otherwise, H_j would not be minimal. So,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Notice that if $r \le H_j/2$, and $2^r \equiv -1 \pmod{p^j}$, then $2^{2r} \equiv 1 \pmod{p^j}$; since H_j is minimal, $H_j \le 2r$. Thus to satisfy both constraints we must have $H_j/2 = r$.

Now assume H_j is odd. Notice that by definition,

$$H_j = \mathrm{ord}_{p^j}(2) = \min\{h > 0 : 2^h \equiv 1 \pmod{p^j}\},$$

so smaller solutions can only be of the form $2^r \equiv -1 \pmod{p^j}$. Then $2^{2r} \equiv 1 \pmod{p^j}$, which means $H_j \mid 2r$, and since H_j is odd, $H_j \mid r$. Thus $H_j \le r$, so H_j is minimal, as desired. ■

We can use this notion of a signed order to find the extension degree of extensions of $\mathbb{F}_2$.

Lemma 4.3. *Let $M > 1$ be odd; let ζ have exact order M; set $\alpha = \zeta + \zeta^{-1}$. Then*

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

Proof. A standard finite field fact says

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min\{r > 0 : \alpha^{2^r} = \alpha\}.$$

By definition of α, $\alpha^{2^r} = \zeta^{2^r} + \zeta^{-2^r}$, so $\alpha^{2^r} = \alpha$ is equivalent to $\zeta^{2^r} + \zeta^{-2^r} = \zeta + \zeta^{-1}$.

For any non-zero $a, b \in \mathbb{F}_2(\alpha)$, $a + a^{-1} = b + b^{-1} \iff (a+b)(ab+1) = 0$. So, either $a = b$ or $a = b^{-1}$. Taking $a = \zeta^{2^r}$ and $b = \zeta$, we obtain

$$\zeta^{2^r} = \zeta \quad \text{or} \quad \zeta^{2^r} = \zeta^{-1}.$$

Since ζ has exact order M, we can equivalently write

$$2^r \equiv 1 \pmod{M} \quad \text{or} \quad 2^r \equiv -1 \pmod{M}.$$

So, $\alpha^{2^r} = \alpha \iff 2^r \equiv \pm 1 \pmod{M}$, and

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min\{r > 0 : \alpha^{2^r} = \alpha\} = \rho_2(M),$$

as desired. ■

Lemma 4.4. *Let a be algebraic over $\mathbb{F}_q$. Then*

$$\mathrm{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(a) = \sum_{r=0}^{[\mathbb{F}_q(a):\mathbb{F}_q]-1} a^{q^r}.$$

Proof. See [3]. ■

Corollary 4.4.1.

$$\mathrm{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(1) = [\mathbb{F}_q(a) : \mathbb{F}_q].$$

Proof. Immediate from evaluating at $a = 1$ in lemma 4.4. ■

Lemma 4.5. *Let $M > 1$ be odd, and let ζ have exact order M. Let $\alpha = \zeta + \zeta^{-1}$. Define*

$$G_M = \{\pm 2^r \pmod{M} : r \ge 0\} \subseteq (\mathbb{Z}/M\mathbb{Z})^\times.$$

Then

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

Proof. Lemma 4.3 showed that

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

So, by lemma 4.4,

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{r=0}^{\rho_2(M)-1} \zeta^{2^r} + \zeta^{-2^r}.$$

By the definition of $\rho_2(M)$, the residues $2^0, -2^0, 2^1, -2^1, \ldots, 2^{\rho_2(M)-1}, 2^{-(\rho_2(M)-1)}$. are all distinct modulo M. Otherwise, we would obtain $2^r \equiv \pm 2^s \pmod{M}$, or equivalently, $2^{r-s} \equiv \pm 1 \pmod{M}$ for some $0 \leq s < r < d$, which contradicts $\rho_2(M)$ being minimal.

The residues are exactly the elements of G_M, so

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u,$$

as desired. ■

Corollary 4.5.1. *If $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, then*

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(M) \pmod{2},$$

where μ is the number-theoretic Möbius function.

Proof. From lemma 4.5, we know that

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

When $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, we also have

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{0 \leq u < M, \gcd(u,M)=1} \zeta^u = \mu(M) \pmod{2},$$

by Ramanujan's sum identity of roots of unity, as desired. ■

Now we may combine together many of our partial results to state a more significant result.

Theorem 4.1. *If p is not a Wieferich prime, then for all $k \in \mathbb{N}$,*

$$d(p^k - 1) = d(p - 1).$$

Proof. Let p be a non-Wieferich prime. The case for $p = 2$ was shown in [5], so we will consider the case when p is odd. For $1 \leq j \leq k$, let

$$R_j = \{\zeta + \zeta^{-1} : \mathrm{ord}(\zeta) = p^j\}.$$

By lemma 1.7,

$$\mathrm{roots}\, F_{p^k} = \bigcup_{j=1}^{k} R_j. \tag{1}$$

Furthermore, since p^k is odd, corollary 1.3.1 and lemma 1.3 tell us that F_{p^k} is the square of a square-free polynomial. In other words, F_{p^k} has $(p^k - 1)/2$ distinct roots. Let $h = \mathrm{ord}_p(2)$. Since p is not Wieferich, corollary 4.1.1 tells us that

$$\mathrm{ord}_{p^j}(2) = \mathrm{ord}_p(2)\, p^{j-1},$$

and lemma 4.2 tells us that

$$\rho_2(p^j) = \rho(p)p^{j-1}.$$

So, by lemma 4.3, if $\alpha \in R_j$,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho(p)p^{j-1}.$$

Suppose both α and $\alpha+1$ are roots of F_{p^k} with $\alpha \in R_j$ and $\alpha + 1 \in R_\ell$. Since $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha+1)$, that is they are the same field with the same degree, α and $\alpha+1$ must both be members of the same R_j and so $j = \ell$.

Suppose for the sake of contradiction that α and $\alpha+1$ are both members of R_j for some $j \geq 2$. Set $M = p^j$ and $m = p^{j-1}$ and choose a primitive Mth root ζ such that $\alpha = \zeta + \zeta^{-1}$. Since $\alpha+1$ is also in R_j, there must exist s such that $\gcd(s,p) = 1$ and $\alpha + 1 = \zeta^s + \zeta^{-s}$. Then

$$1 + \alpha + (\alpha+1) = 1 + \zeta + \zeta^{-1} + \zeta^s + \zeta^{-1} = 0.$$

Equivalently, ζ is a root of

$$P_s(x) = 1 + x + x^{M-1} + x^s + x^{M-s}.$$

Notice that ζ^m is a primitive pth root. Let $f(x)$ be the minimal polynomial of ζ^m over $\mathbb{F}_2$. The degree of $f(x)$ is $\text{ord}_p(2)$. Since $f(\zeta^m) = 0$, ζ is a root of $f(x^m)$. The degree of $f(x^m)$ is $\text{ord}_p(2)\, m = \text{ord}_{p^j}(2)$. This is also the degree of the minimal polynomial of ζ and so

$$f(x^m) \mid P_s(x).$$

So, every polynomial can be decomposed by exponents modulo M:

$$P_s(x) = \sum_{a=0}^{M-1} X^a P_a(X^M). \tag{2}$$

In P_{s-1} the only exponent dividing M is 0, so $P_0(X^M) = 1$. But if $f(x^m) \mid P_s(x)$, then $f(x)$ must divide every compoenent $P_a(x)$, including $P_0(x) = 1$. This is impossible because f isn't constant. Thus we have arrived at a contradiction and must conclude that no common roots of $F_{p^k}(x)$ and $F_{p^k}(x+1)$ are in R_j for $j \geq 2$.

So the only common roots of $F_{p^k}(x)$ and $F_{p^k}(x+1)$ must come from R_1 and so we must conclude

$$d(p^k - 1) = d(p-1).$$

■

We can combine this result with our recurrence when doubling n and adding 1.

Corollary 4.5.2. *Let p be an odd non-Wieferich prime. Then for all $n, m \geq 0$,*

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p-1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

Proof. The $m = 0$ case reduces to what Sutner already showed [5]. For the other cases, we first apply corollary 3.1.1 to obtain

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p^m - 1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

We then apply theorem 4.1 to change $d(p^m - 1)$ to $d(p-1)$. ■

Corollary 4.5.3. *For all $n, m \geq 1$, the following relationships hold*

1. $d(2^n - 1) = 0$

2. $d(3^n - 1) = 0$
3. $d(5^n - 1) = 4$
4. $d(7^n - 1) = 0$
5. $d(11^n - 1) = 0$
6. $d(13^n - 1) = 0$
7. $d(17^n - 1) = 8$
8. $d(19^n - 1) = 0$
9. $d(23^n - 1) = 0$
10. $d(29^n - 1) = 0$
11. $d(31^n - 1) = 0$
12. $d(37^n - 1) = 0$
13. $d(41^n - 1) = 0$
14. $d(43^n - 1) = 0$
15. $d(2^n 3^m - 1) = 2^{n+1} - 2$
16. $d(2^n 5^m - 1) = 2^{n+2}$
17. $d(2^n 7^m - 1) = 0$

You may have noticed that for many sizes, we have $d(p-1) = 0$. We can actually classify some p for which this is the case.

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